

LAB 7 – Encapsulation

1. Encapsulation with full properties (backing field)

```
class BankAccount
{
    private double Balance;
    public double Balance1
    {
        get { return Balance; }
        set{
            if (value<0)
            {
                Console.WriteLine("Balance cannot be negative");
            }
            else
            {
                Balance = value;
            }
        }
    }
}
```

```
class Program
{
    static void Main(string[] args)
    {
        BankAccount b1 = new BankAccount();
        b1.Balance1 = 100;
        Console.WriteLine($"Your balance is {b1.Balance1}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run
Your balance is 100
```

2. Encapsulation with auto – implemented properties

```
class Student
{
    public string Name { get; set; }
    public int RollNo { get; set; }
    public double Marks { get; set; }
}
```

```
class Program
{
    static void Main(string[] args)
    {
        Student s1 = new Student();
        s1.Name = "Rahul";
        s1.RollNo = 101;
        s1.Marks = 90.5;
        Console.WriteLine($"Name: {s1.Name}, RollNo: {s1.RollNo}, Marks: {s1.Marks}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run  
Name: Rahul, RollNo: 101, Marks: 90.5
```